Milestone 2 Team Plan — What Makes a Song Popular?
Milestone 2 小組計畫 — 什麼讓一首歌受歡迎？

PBA Fall 2026 · Group 13 · 張宏敏 (leader 組長), 陳禹之, 卓妤薪, 黃湘琴, 薛函書
Repo: https://github.com/MichaelCHM0910/pba-project-group-13 · Prepared 2026-10-24

> **Deadline: Sun 25 Oct 2026, 23:59.** Only the leader submits, as a PDF on eeclass. Please read this and leave comments by **Sun 25 Oct, 18:00**.
> **截止：2026/10/25（日）23:59**，由組長在 eeclass 繳交 PDF。請大家在 **10/25 18:00 前**看完並回覆意見。

1. What Milestone 2 asks for ｜ 作業要求

English	中文
A 1–2 page proposal (informal is fine) that covers: 1. The research question in one sentence, and whether it is descriptive, predictive, or causal. 2. The hypothesis and why it might be true (a plausible mechanism matters more than a citation). 3. A data description: source and how it was collected, unit of observation, period, number of observations, and a variable table (name, meaning, unit). 4. The key variables: explanatory variable / treatment and outcome, and how each is measured. 5. What would count as evidence: which pattern supports the hypothesis and which contradicts it. Data must be real (not synthetic Kaggle data); a Kaggle dataset is OK only if its original source is documented. Load the data in R before writing the proposal. Nothing is locked in; we can change later.	一份 1–2 頁的提案（不用太正式），內容包括： 1. 一句話的研究問題，並說明是描述型、預測型還是因果型。 2. 假設，以及為什麼可能成立（合理的機制比引用文獻重要）。 3. 資料描述：來源與蒐集方式、觀察單位、期間、樣本數，以及變數表（名稱、意義、單位）。 4. 關鍵變數：解釋變數（或處理）與結果變數，各自怎麼衡量。 5. 什麼算證據：什麼樣的資料型態支持假設、什麼樣的型態推翻假設。 資料必須是真實資料（不能用 Kaggle 上的合成資料）；Kaggle 資料要能查到原始來源才可以用。寫提案前要先在 R 讀進資料檢查。提案不是定案，之後可以改。

2. Proposed question ｜ 研究問題

English	中文
Which audio features of a song (danceability, energy, loudness, length, and others) predict its popularity on Spotify, after accounting for its genre and release year? Type: **predictive**, with a descriptive first part. We do not claim causation: songs are not randomly assigned their features, and artist fame is not in the data.	**在控制曲風與發行年份之後，一首歌的音訊特徵（好不好跳、能量、音量、長度等）能不能預測它在 Spotify 上的熱門程度？** 類型：**預測型**，先做一段描述型分析。我們不主張因果：歌曲的特徵不是隨機分配的，資料裡也沒有歌手知名度。

3. Hypotheses ｜ 假設與理由

English	中文
H1. Danceable, loud, and shorter songs are more popular. Streaming pays per play and a play counts after 30 seconds, so shorter songs are replayed more and skipped less; danceable, loud songs fit the party, workout, and "hits" playlists that drive most listening.	**H1.** 好跳、大聲、比較短的歌比較紅。串流以「播放次數」計酬，聽超過 30 秒就算一次播放，所以短歌會被重播更多、被跳過更少；好跳又大聲的歌適合派對、健身、熱門歌單，而這些歌單貢獻了大部分的收聽量。
H2. Instrumental songs are less popular, because vocals and lyrics are what listeners remember, sing, and share.	**H2.** 純音樂（沒有人聲）比較不紅，因為人聲和歌詞才是聽眾記得、會跟著唱、會分享的部分。

H3. Genre and release year matter more than audio features. Spotify's score is based on recent plays, so new releases score higher, and pop has a larger audience than EDM.	**H3.** 曲風和發行年份的影響比音訊特徵大。Spotify 的分數主要看「最近」的播放量，所以新歌分數較高；流行樂的聽眾也比電子舞曲多。

4. Data ｜ 資料描述

English	中文
Source: "Spotify Songs", TidyTuesday 2020-01-21. Collected in Jan 2020 by Kaylin Pavlik from the official Spotify Web API with the `spotifyr` R package, by pulling songs from playlists in six genres. Comes with a codebook. It is not synthetic Kaggle data. **Unit:** one song. **Period:** popularity snapshot from Jan 2020; release years 1957–2020 (median 2016). **Size:** 32,833 rows (song × playlist) → 28,356 unique songs. No missing values in the variables we use. Six genres, 4,100–5,400 songs each. 7.6 MB, so it fits on GitHub.	**來源**：TidyTuesday 2020-01-21 的「Spotify Songs」。由 Kaylin Pavlik 在 2020 年 1 月用 R 的 `spotifyr` 套件從 Spotify 官方 API 抓取，來自六種曲風的歌單，附有 codebook。不是 Kaggle 的合成資料。 **觀察單位**：一首歌。**期間**：熱門度是 2020 年 1 月的快照；發行年份 1957–2020（中位數 2016）。 **樣本數**：32,833 列（歌 × 歌單），去除重複後 28,356 首歌。我們要用的變數沒有缺值。六種曲風各 4,100–5,400 首。檔案 7.6 MB，可以放 GitHub。

Variable 變數	Meaning	意義	Unit 單位	Role 角色
track_popularity	Spotify popularity, mostly recent plays	Spotify 熱門度，主要依近期播放量	0–100	Outcome Y 結果
danceability	Suitability for dancing	適合跳舞的程度	0–1	X
energy	Perceived intensity	強度、激烈程度	0–1	X
loudness	Average loudness	平均音量	dB	X
duration_min	Song length	歌曲長度	minutes 分鐘	X
instrumentalness	Likelihood of no vocals	沒有人聲的可能性	0–1	X
valence	Positiveness (happy vs. sad)	情緒正向程度（快樂/悲傷）	0–1	X
acousticness / speechiness / liveness	Acoustic; spoken words; live	原聲樂器；口說成分；現場錄音	0–1	X
tempo	Tempo	速度	BPM	X
mode	Major or minor key	大調或小調	major/minor	X
genre	Playlist genre	歌單曲風	6 categories 類	Control 控制
release_year	Album release year	專輯發行年份	year 年	Control 控制

5. Evidence and first look ｜ 什麼算證據、初步結果

English	中文
Supports: holding genre and year fixed, danceability and loudness have positive coefficients, duration and instrumentalness negative; and adding audio features lowers test-set prediction error (RMSE) compared with a genre + year model. **Contradicts:** coefficients near zero or of the opposite sign, or no gain in test-set accuracy, meaning popularity depends on things outside the sound (artist, marketing, playlists).	**支持假設**：在控制曲風和年份後，danceability、loudness 係數為正，duration、instrumentalness 係數為負；且加入音訊特徵後，測試集的預測誤差（RMSE）比只用曲風 + 年份的模型低。 **推翻假設**：係數接近 0 或方向相反，或加入音訊特徵後測試集準確度沒有提升，代表熱門度取決於聲音以外的因素（歌手、行銷、歌單）。

First check (already done): correlations with popularity are weak: duration −0.14, instrumentalness −0.12, energy −0.10, danceability +0.05. Mean popularity: pop 45.9, rap 41.8, Latin 41.4, rock 39.7, R&B 35.9, EDM 30.7. This hints H3 may hold, and energy goes against H1. That is fine: an honest "audio features matter little" is a valid result.	**初步檢查（已做）：**各特徵與熱門度的相關都很弱：長度 −0.14、純音樂程度 −0.12、能量 −0.10、好跳程度 +0.05。各曲風平均熱門度：pop 45.9、rap 41.8、Latin 41.4、rock 39.7、R&B 35.9、EDM 30.7。看起來 H3 可能成立，而能量的方向和 H1 相反。這沒關係：誠實地得出「音訊特徵影響不大」也是有效的結論。

6. Methods and limitations ｜ 方法與限制

English	中文
Methods: descriptive plots by genre and year; linear regression with controls; train/test split to compare prediction models; logistic regression for "hit" (popularity ≥ 70). **Robustness:** repeat without the 9% of songs with popularity 0, and within each genre. **Limitations:** songs come from playlists, not a random sample; popularity is one snapshot; artist fame is missing. No personal data.	**方法：**依曲風、年份做描述圖表；加控制變數的線性迴歸；切訓練集/測試集比較預測模型；用 logistic regression 預測「熱門歌」（熱門度 ≥ 70）。 **穩健性：**排除 9% 熱門度為 0 的歌（多是重複發行的版本）再跑一次，並分曲風各跑一次。 **限制：**歌曲來自歌單而非隨機抽樣；熱門度只是某一時點的快照；沒有歌手知名度。資料不含個人資訊。

7. What is already on GitHub ｜ GitHub 上已完成的部分

English

- `R/01_get_data.R`: downloads the raw data into `data/raw/` (not uploaded) and saves the cleaned file `data/spotify_clean.csv` (one row per song).
- `data/codebook.md`: variable table.
- `analysis/01_data_check.qmd`: sample size, missing values, popularity histogram, popularity by genre and year, correlations, a first regression.
- `proposal/`: the English proposal to submit, and this file.

To run: in RStudio, **Pull** → open the project → run `R/01_get_data.R` → Render `analysis/01_data_check.qmd`. Needs packages `tidyverse` and `here`.

中文

- `R/01_get_data.R`：下載原始資料到 `data/raw/`（不上傳），清理後存成 `data/spotify_clean.csv`（一首歌一列）。
- `data/codebook.md`：變數表。
- `analysis/01_data_check.qmd`：樣本數、缺值、熱門度分布圖、各曲風與年份的熱門度、相關係數、第一個迴歸。
- `proposal/`：要繳交的英文提案，以及這份檔案。

執行方式：在 RStudio 先 **Pull** → 開啟專案 → 執行 `R/01_get_data.R` → Render `analysis/01_data_check.qmd`。需要套件 `tidyverse` 和 `here`。

8. Suggested roles (please claim one) ｜ 建議分工（請認領）

Role 角色	Tasks	工作內容	Owner 負責人
Coordinator 統籌	Repo, data script, merge, submit milestones	管理 repo、資料程式、整合、繳交	張宏敏
Descriptive 描述分析	Plots of popularity by genre, year, features	熱門度依曲風、年份、特徵的圖表	
Regression 迴歸	Linear models with controls; interpret coefficients	加控制變數的迴歸、解讀係數	
Prediction 預測	Train/test split, RMSE, "hit" logistic model	訓練/測試集、RMSE、熱門歌 logistic 模型	
Writing 撰寫	Background, mechanisms, limitations, report editing	背景、機制、限制、報告潤稿	

9. Before the deadline ｜ 截止前每個人要做的事

English	中文

1. Read the English proposal (`proposal/milestone2_proposal.pdf`) and reply with changes by Sun 18:00.
2. Claim a role in section 8.
3. Optional: Pull and run the data check, to make sure the code works on your computer.

1. 閱讀英文提案（`proposal/milestone2_proposal.pdf`），週日 18:00 前回覆修改意見。
2. 在第 8 節認領一個角色。
3. 選做：Pull 下來跑一次資料檢查，確認程式在自己電腦上能跑。